

WOLAITA SODDO UNIVERSITY
COLLEGE OF NATURAL AND COMPUTATIONAL
SCIENCES
DEPARTMENT OF PHYSICS
General Physics (Phys 101) Mid. Term Exam for fresh man
students (Natural science stream)

Time Allotted: 2:00 Hours
Max Marks: 30%
Program: Regular

Course Code: Phys1011
Date of exam 29/06/2022 E.C.
Semester: 1

Name KIRUBEL ARSE Id. No. 11111/69603/14 Section 32

Part I Multiple choose 20% (1 pt. each)

Choose the correct answer from the given alternatives and write your appropriate answer on the answer sheet

Unit vector along the vector $4\mathbf{i} + 3\mathbf{j}$ is _____
A. $(4\mathbf{i} + 3\mathbf{j})/5$ C. $4\mathbf{i} + 3\mathbf{j}$
B. $(4\mathbf{i} + 3\mathbf{j})/6$ D. $(4\mathbf{i} + 3\mathbf{j})/10$

Under what condition is average velocity equal to the average of the object's initial and final velocity?

- A. This can only occur if there is no acceleration.
- B. The acceleration is constant.
- C. This can occur only when the velocity is zero.
- D. The acceleration must be constantly increasing

10-kg rock and 20-kg rock are dropped from the same height and experience no significant resistance. If it takes the 20-kg rock a time T to reach the ground, what time will it take the 10-kg rock to reach the ground?

- A. $4T$ C. $2T$
- B. T D. $T/2$

A car initially traveling at 60 km/h accelerates at a constant rate of 2.0 m/s^2 . How much time is required for the car to reach a speed of 90 km/h?

- A. 15 s C. 30 s
- B. 45 s D. 4.2 s





5. If an object is thrown vertically up with the initial speed u from the ground, then the time taken by the object to return back to ground is
- A. $u^2/2g$ C. u^2/g
B. $u/2g$ D. $2u/g$
6. A body of mass 10 kg, projected at an angle of 30° from the ground with an initial velocity of 5 m/s, acceleration due to gravity is $g = 10 \text{ m/s}^2$, what is the time of flight?
- A. 0.866 s C. 1.86 s
B. 1.96 s D. 1.862 s
7. When do we get maximum height in a simple projectile motion?
- A. When $\theta = 45^\circ$ C. When $\theta = 60^\circ$
B. When $\theta = 90^\circ$ D. When $\theta = 0^\circ$
8. A body of mass 10 kg, is moving with a velocity of 5 m/s in a circle of radius 5 m, what is the centripetal acceleration of the body?
- A. 5 m/s^2 C. 25 m/s^2
B. 0.5 m/s^2 D. 50 m/s^2
9. Kepler's 2nd law deals with
- A. The shape of the planet's orbits
B. The speed/area the planet travels
C. The length of time it takes the planet to orbit the sun
D. Time travel
10. When a force is parallel to the direction of motion of the body, then work done on the body is.
- A. Zero C. minimum
B. Infinity D. maximum

Which of the following is true?

- A. If a car is traveling east ward, its acceleration is eastward.
B. If a car is slowing down, its acceleration must be negative.
C. A particle with constant acceleration can never stop and stay stopped.
D. All can be the possible answer
- the position of a particle as function of time from the origin is given by $x(t) = 5t^2$. What the magnitude of the acceleration of the stone at $t = 2\text{s}$?
- A. 10m/s B. 20m/s C. 40m/s D. 25m/s

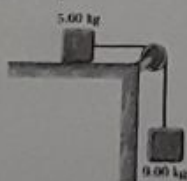
20. If a body starts from a point and returns back to the same point, then it is

- A. Average velocity is zero but not average speed
- B. Average speed is zero, but not average velocity
- C. Both average speed and velocity are zero
- D. Average speed and velocity depends upon the path.

Part II work out (10%) 2.5pts. each

Show all the necessary steps neatly and clearly

1. A particle initially located at the origin has an acceleration of $a = 3.00j \text{ m/s}^2$ and an initial velocity of $v_i = 500i \text{ m/s}$. Find
 - (a) The vector position and velocity at any time t and
 - (b) The coordinates and speed of the particle at $t = 2.00 \text{ s}$.
2. A force $\vec{F} = 6i + 2j \text{ N}$ acts on a particle that undergoes a displacement $\Delta\vec{r} = (3i + j) \text{ m}$. Find (a) the work done by the force on the particle and (b) the angle between $\vec{F}$ and $\Delta\vec{r}$.
3. A 5.00-kg object placed on a frictionless, horizontal table is connected to a string that passes over a pulley and then is fastened to a hanging 9.00-kg object, as in Figure below.
 - a. Draw free-body diagrams of both objects.
 - b. Find the acceleration of the two objects and
 - c. The tension in the string.



4. An older model car accelerates from rest to speed $2v$ in 5 seconds. A newer, more powerful car accelerates from rest to 3 times the speed of old car within 5 seconds. What is the ratio of the power of the newer car to that of the older car?